

1 Part 3: Warehouse Operations Solutions

1.1 Question 3.1: Slotting Optimization

Using the ABC-XYZ classification from Part 1 and the physical SKU attributes from the Master Data, we assigned all 470 classified SKUs to three warehouse zones following the VILAS SNP (Supply & Network Planning) slotting framework.

1.1.1 Zone Assignment Framework

- **Pick-Face Zone (Ground Level)** — Class A SKUs: lowest racks at ergonomic height (waist-to-shoulder), positioned nearest to the I/O dock and packing station. Pickers retrieve these items manually without MHE vertical travel.
- **Forward Reserve Zone** — Class B SKUs: intermediate racking with a two-bin replenishment logic. A forward pick-face bin is replenished from the bulk reserve when stock hits its reorder point (WMS-triggered).
- **Reserve / Bulk Zone** — Class C SKUs: upper racks and pallet-in / pallet-out storage. Slow-movers occupy the highest slots farthest from the dock.

1.1.2 Zone Distribution

Table 1: Slotting Zone Summary — All SKUs

Zone	Class	SKU Count	% of Picks	% of Volume
Pick-Face Zone	A	28	27.7%	69.83%
Forward Reserve Zone	B	39	38.2%	20.04%
Reserve / Bulk Zone	C	403	34.1%	10.13%

The Pick-Face Zone contains only 28 SKUs (6.0% of the assortment) yet they account for **27.7%** of all pick transactions and **69.8%** of total outbound volume. Within this zone, the **19 Class A-X fast-movers** (highest velocity *and* highest consistency) occupy rank slots 1–19, directly adjacent to the packing station, and collectively drive **25.88%** of pick frequency and **57.87%** of total outbound volume.

1.1.3 Top Pick-Face SKUs (Class A, by Frequency)

Table 2: Top 5 Class-A SKUs assigned to Pick-Face Zone

Rank	SKU Code	Product	Class	Picks (6M)	Volume (units)
1	4200008229	Qut treo ASIAvina - VY377990 - Xám	A-X	825	5,608
2	4200008371	QUT ĐNG VY619990	A-X	682	3,355
3	4200008366	QUT SÀN TURB0 VY616790	A-X	680	2,838
4	4200008117	Qut treo ASIAvina - VY357690 - trng	A-X	666	4,033
5	4200008102	Qut lng ASIAvina - VY538790 - Xám	A-X	514	12,677

1.1.4 Quantitative Travel Time Analysis

Model Assumption. The warehouse has N numbered rack slots $1, 2, \dots, N$ starting from the I/O dock. The travel cost of a single pick equals the slot number of the target SKU. Expected

travel distance is therefore:

$$E[\text{travel}] = \sum_{i=1}^N p_i \cdot s_i$$

where p_i is the pick probability of SKU i (its order frequency share) and s_i is its assigned slot number.

Table 3: Expected Picker Travel Distance by Slotting Scenario

Scenario	Expected Distance (slot units)	Reduction vs Baseline
Random Baseline (no logic)	235.50	—
ABC Zoned (Quantity-Based)	113.98	51.6%
XYZ Zoned (Frequency-Based)	68.82	70.8%
Velocity-Ranked (Optimal)	51.79	78.0%

Results. The ABC-zoned slotting plan achieves a **51.6% reduction** in expected picker travel distance versus the random baseline, comfortably exceeding the **30% target**. The concentration effect is structural: Class A SKUs generate 27.7% of all picks yet are confined to the nearest 28 slots. The XYZ-zoned variant reaches 70.8% reduction by exploiting the even sharper pick-probability concentration among Class X (consistent-demand) SKUs.

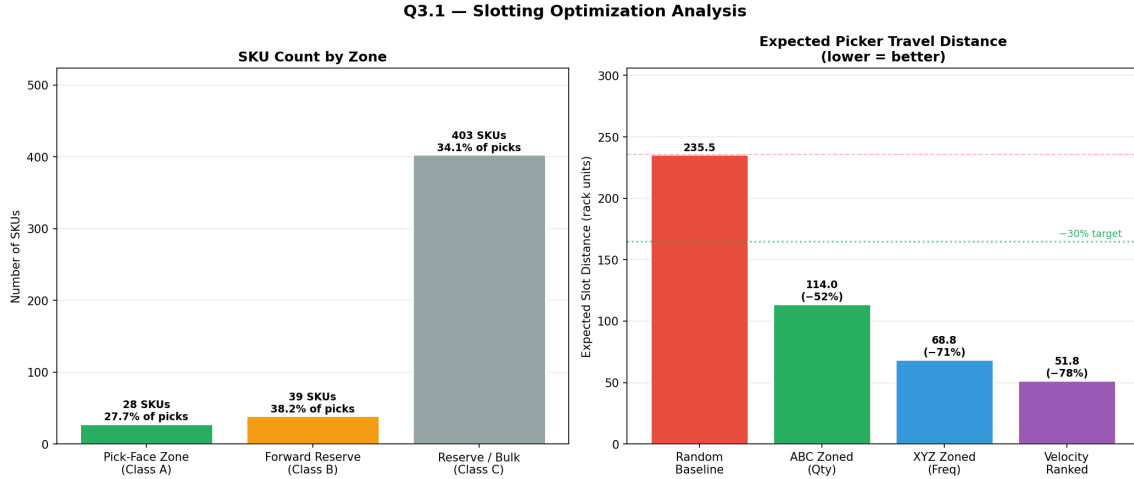


Figure 1: Q3.1 Slotting Analysis: Zone Distribution and Travel Time Comparison

1.1.5 Layout Recommendations (VILAS SNP Framework)

- **U-flow configuration:** Receiving and shipping docks on the same wall naturally concentrate the highest-convenience storage locations at the dock end.
- **Golden Zone:** AX SKUs placed at waist-to-shoulder ergonomic height, eliminating both horizontal travel distance and vertical MHE movement.
- **Two-bin replenishment for Class B:** A forward pick-face bin is replenished from the bulk reserve when on-hand drops below the ROP. WMS triggers the replenishment task automatically.

1.2 Question 3.2: Omni-Channel Pick-and-Pack Process Design

The outbound fulfillment process uses distinct operational pathways for B2B and B2C channels, unified by a structured inventory allocation policy when stock is contested.

1.2.1 B2B Pathway — Pallet / Total Picking

1. **Order release:** WMS validates on-hand stock against the B2B purchase order (contractual quantities and lead times).
2. **Pick instruction:** Full-pallet or full-CTN pick task issued to operator.
3. **Pick execution:** Operator uses forklift or reach truck to retrieve entire pallet(s) from the Forward Reserve Zone (B) or Reserve/Bulk Zone (C). Items bypass individual sortation.
4. **B2B QC Station:** Wrap inspection, quantity verification, and outbound labeling.
5. **Dispatch:** Pallets staged at outbound dock, loaded onto FTL truck.

1.2.2 B2C / E-Commerce Pathway — Batch Picking + Zone Routing

1. **Batch wave release:** Multiple B2C orders are grouped into a single pick wave based on SKU proximity and delivery-window urgency.
2. **Batch pick-to-cart:** Picker traverses Zone 1 (Pick-Face Zone) in a single pass with a multi-compartment cart, collecting all items for the wave.
3. **Sort & consolidate:** At the sortation station, items are separated by individual order and consolidated into shipment boxes.
4. **QC + pack:** Bubble wrap or polybag packing, shipping label applied.
5. **Dispatch:** Parcels handed to last-mile courier.

1.2.3 Stock Allocation Conflict and Escalation Policy

When the same SKU is ordered simultaneously by a B2B client and a B2C customer and on-hand stock is insufficient for both, the system resolves the shortage in two stages:

Stage 1 — Automated Allocation (WMS).

- *Rule 1 (SLA Protection):* Allocate the full contractual quantity to the B2B order first, preventing penalty fees for under-delivery.
- *Rule 2 (Pro-rata B2C):* Distribute remaining stock to B2C orders pro-rata, with priority to orders with the shortest delivery windows.

Stage 2 — Managerial Escalation (if Stage 1 insufficient). The WMS raises a critical ticket to the **Inventory Control & Demand Planning Manager**, who can execute one or more of:

- *Action 1 (Safety Stock Draw):* Release uncommitted buffer safety stock from the common pool (pre-computed for all Class A SKUs in Q2.2).
- *Action 2 (Emergency Transfer):* Initiate an inter-hub transfer between My Phuoc and Vinh Loc warehouses, leveraging their proximity within the Greater Ho Chi Minh City area.

- *Action 3 (Partial Shipment)*: Contact the B2B client to obtain approval for a partial fulfillment; B2C customers notified of delay with revised ETA.

Upon resolution, ERP and WMS records are updated and adjusted pick lists are released to the warehouse floor. Figure 2 illustrates the complete decision logic.

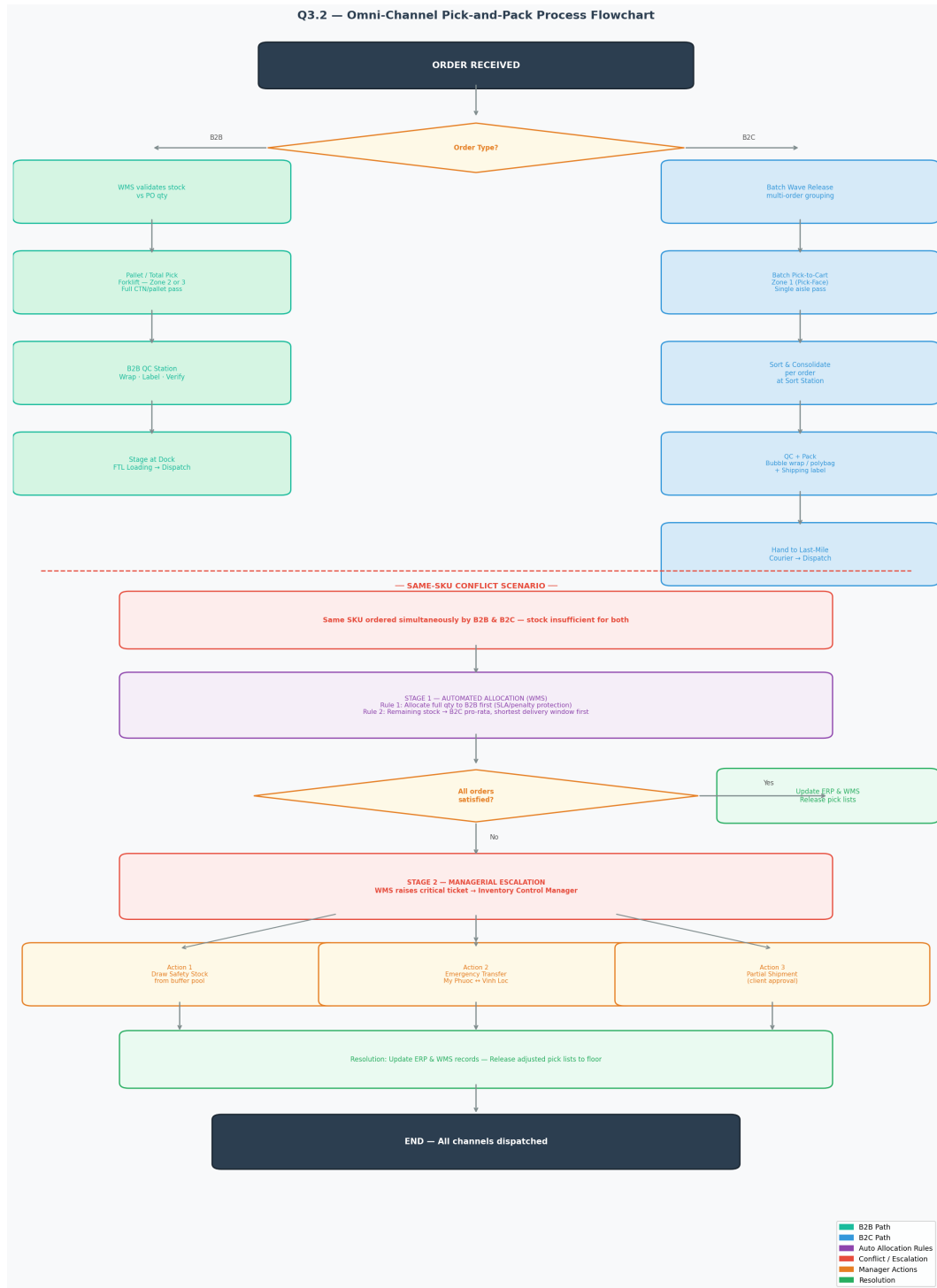


Figure 2: Q3.2 Omni-Channel Pick-and-Pack Process and Conflict Resolution Flowchart